University of Central Punjab LAHORE

Assignment # 2

Total Marks: _30 Submission Date: 03/04/2026 before 11 a.m

Name: **Roll No:** **Class:** Database D 5

I. Assignment should be hand written.

II. You should create your own solution according to your logic. Any similarity with other student and internet will be consider as cheating and your assignment will be marked as ZERO.

III. Attach this page as front page of your assignment solution. IV. Submission should be online within time and date.

Question#1: Consider Given Tables (20 Marks)

1. ACTOR (Act_id, Act_Name, Act_Gender)
2. DIRECTOR (Dir_id, Dir_Name, Dir_Phone)
3. MOVIES (Mov_id, Mov_Title, Mov_Year, Mov_Lang, Dir_id)
4. MOVIE_CAST (Act_id, Mov_id, Role)
5. RATING (Mov_id, Rev_Stars)

Write Queries for Given Scenarios

1. Create a schema with SQL Queries. (Along all Constraints, PK, FK and Referential Constraint if it is necessary for database existence and maintenance etc)
2. Insert at-least 30 Rows in each table
3. List the titles of all movies directed by 'Hitchcock'.
4. Show all movies with their cast.
5. Show how many actor perform 'brother' Roles in any movie.
6. Find the movie names where one or more actors acted in two or more movies.
7. List all actors who acted in a movie before 2021 and also in a movie after 2023.
8. Find movies in which actor is also director of the movie.
9. Show all movies director who directed more than 5 movies from 2020 to 2024.
10. Find the title of movies and number of stars for each movie that has at least one rating and find the highest number of stars that movie received. Sort the result by movie title.
11. Calculate the rating of all movies.
12. Show how many movies were directed by 'Steven Spielberg'.

Assignment # 2
Database System
Question # 1

SQL:

Create database if not exists Assignment_02;

Use Assignment_02;

Select DATABASE();

Create table Actor (act_id INT NOT NULL AUTO_Increment,
act_name VARCHAR(50) NOT NULL, act_gender CHAR(1)
NOT NULL CHECK (act_gender IN ('M', 'F')),
Primary key (act_id));

2. Director table

Create table Director (dir_id INT NOT NULL AUTO_INCREMENT,
dir_name VARCHAR(50) NOT NULL, dir_Phone VARCHAR(15)
UNIQUE, Primary Key (dir_id));

3. Movie table

Create table Movie(
mov_id INT NOT NULL AUTO_INCREMENT,
mov_title VARCHAR(50) NOT NULL,
mov_year INT NOT NULL CHECK (mov_year >=1900 AND
mov_year <= 2030),
mov_lang VARCHAR(20) not null Default 'English',
mov_type VARCHAR (20),
dir_id INT NOT NULL,
Primary Key (mov_id),

constraint fk_movie_director FOREIGN KEY (dir_id)
References Director (dir_id)
ON delete cascade ON update Cascade);

4. Movie Cast table

```
Create table Movie_cast(
cast_Role VARCHAR(50) NOT NULL,
act_id INT NOT NULL,
mov_id INT NOT NULL,
Primary key (act_id, mov_id, cast_role),
Constraint fk_cast_actor FOREIGN KEY (act_id)
    References Actor (act_id)
ON delete cascade ON update cascade,
Constraint fk_cast_movie FOREIGN KEY (mov_id)
    Refrences Movie (mov_id)
ON delete cascade ON update cascade );
```

5. Rating table

```
Create table Rating (
rating_id int Auto_increment, mov_id INT,
rev_stars Decimal (2,1) CHECK (rev_stars >=0 AND rev_stars <=5.0),
Primary key (rating_id),
Constraint fk_rating_movie FOREIGN KEY (mov_id)
 Refrences Movie (mov_id) );
```

```
INSERT INTO ACTOR(act_name, act_gender) VALUES
('Cary Grant', 'M'),('Grace Kelly', 'F'),('James Stewert','M'),('Kim Novak','F'),
('Trippi Harden','F'),('Tom Hanks', 'M'),('Meryl Streep','F'),('Meg Ryan','F'),
('Harrison Ford','M'),('Richard Dreyfuss','M'),('Denzel Washigton','M'),
('Sandra Bullock','F'),('Bradd Pitt','M'),('Angelina Jolie','F'),('Leonardo Cappric','M'),
('Kate Winslot','F'),('Johney Depp','M'),('Helna Carter','F'),('Will Smith','M'),
('Cameron Diaz','F'),('Kaneu Reeves','M'),('Jennifer Aniston',F),
('Robert Downey Jr','M'),('Scarllet Johansson','F'),('Chris Evans','M'),
('Cate Blanchett','F'),('Sam Neill','M'),('Laura Denn','F'),('Joseph Dern','M')
,('Embeth Dave','F');
```

```
INSERT INTO Director(dir_name, dir_phone) VALUES
('Alfred Hichcock','03001000001'),('steven Speilberg','03001000002),(Martin Scorsese',
'03001000003'),('Quentin Tarintino','03001000004'),('Chris Nolan',0300-1000005'),
('James cameron','03001000006'),('Ridley Scott','03001000007'),('David Fincher','03001000008'),
('Denis Vellenue','03001000009'),('Wes Anderson','03001000010'),('Joel Coen',
'03001000011'),('Ethan Coen','03001000012'),('Paul Thomas','03001000013'),
('Daren Aronfsky','03001000014'),('Duncan Jones','03001000015'),
('Michel Gondry','03001000016'),('Charlie Kaufman','03001000017'),
('Tony Scott','03001000018'),('Michael Mann','03001000019'),
('spike Lee','03001000020),('Katherine Beglow','03001000021'),
('Peter Jackson','03001000022'),('Franciss cappola','03001000023'),
('Stanley Kubrick','03001000024'),('George Lucas','03001000025'),
('Satyajit Ray','03001000026'),('Akira Kurosawa','03001000027'),
('Roman Polanski','03001000028'),('Elia Kazan',03001000029'),
('Billy Wilder','03001000030');
```

Movie table

Insert into movies (mov_title, mov_year, mov_lang, mov_type, dir_id) values ('Vertigo', 1958, 'English', 'Thriller', 1),
('Rear Window', 1954, 'English', 'Thriller', 1), ('Psycho', 1960,
'English', 'Horror', 1), ('The Birds', 1963, 'English', 'Thriller', 2),
('Rope', 1948, 'English', 'Thriller', 1), ('Jaws', 1975, 'English', 'Thriller', 2),
('Jurassic Park', 1993, 'English', 'Action', 2), ('E.T the Extra', 1982, 'English', 'Sci'
2), ('Warhorse', 2011, 'English', 'Adventure', 2), ('Saving Private', 1998, 'English', 'War',
'Drama' 1999 1980
2), ('Taxidriver', 'English', 2005, 3), ('Raging', 'English', 'Drama', 3), ('The Man', 'English',
1997
'Crime', 3), ('The Wolf', 'English', 'Drama', 3), ('Pulp fiction', 'English', 1994, 'Crime', 4),
('Kill Bill', 2003, 'English', 'Action', 4), ('Inglories', 2009, 'English', 'War', 4),
('Hateful', 2015, 'English', 'Drama', 5), ('Django', 2012, 'English', 'western', 5), ('Inception', 2010,
'English', 'War', 5), ('Interseller', 2009, 'English', 'Drama', 5), ('Dark Knight', 2008, 'English', 'Sci', 6)
('Titanic', 1997, 'English', 'Romance', 6), ('Avatar', 2009, 'English', 'Sci', 6), ('Gladiator', 2000,
'Action', 7) ('Blade Runner', 1997, 'English', 'War', 7), ('Social', 2005, 'english', 'Sci', 7), ('Gone', 1990, 'english',
'Drama', 8), ('Dune', 2021, 'english', 'Sci', 9);

Movie cast table

Insert into Movie_cast (cast_role, act_id, mov_id) VALUES ('hero', 1, 1),
('heroine', 2, 1), ('brother', 3, 1), ('friend', 4, 1), ('hero', 3, 2), ('heroine', 4, 3), ('brother', 3, 2)
('friend', 5, 2) ('hero', 1, 3), ('heroine', 2, 2), ('brother', 3, 3), ('friend', 4, 3), ('hero', 5, 4)
('heroine', 2, 4), ('brother', 1, 4), ('friend', 5, 5), ('hero', 8, 7) ('heroine', 8, 8) ('brother', 8, 8)
('friend', 5, 8), ('heroine', 8, 9), ('hero', 9, 8), ('brother', 27, 8), ('friend', 27, 8), ('hero', 20, 9)
('brother', 18, 10), ('hero', 20, 20), ('heroine', 22, 21), ('friend', 25, 9), ('hero', 9, 20),
('heroine', 29, 25);

Question # 6

```
Select mov-title from Movie
Where mov-id IN (Select mov-id from Movie-cast
    Where act-id IN (Select act-id from Movie-cast
    Group by act-id Having Count (mov-id >= 2));
```

Question # 7

```
Select act_name from Actor
Where act-id IN (Select act-id from Movie_cast
    Where mov-id IN (Select mov-id from Movie Where
    mov-year < 2021)) AND act-id IN (Select act-id
      from Movie-cast Where mov-id IN (Select
       (mov-id from Movie Where mov-year > 2023));
```

Question # 8

```
Select mov-title from Movie
Where mov-id IN (Select mov-id From Movie-cast
    Where act-id IN (Select act-id from
    Actor Where act-name IN (Select
     dir-name from Director)));
```

Question-09

```
SELECT dir_name FROM Director
Where dir_id IN(SELECT dir_id FROM MOVIE WHERE
(mov_year >= 2020 AND mov_year <= 2024)
Group by dir_id HAVING COUNT(mov_id)>5);
```

Question-10

```
SELECT mov_title, (SELECT MAX(rev_stars) FROM Rating
Where Rating.mov_id=Movie.mov_id) AS highest_stars
FROM Movie
WHERE mov_id (SELECT mov_id FROM Rating)
ORDER by mov_title;
```

Question-11

```
SELECT (SELECT mov_title FROM Movie WHERE Movie.mov_id
= Rating.mov_id) AS mov_title, AVG(rev_stars) As avg-rating
FROM Rating Group by mov_id;
```

Question-12

```
SELECT mov_title from Movie
WHERE dir_id =(SELECT dir_id FROM Director WHERE
 dir_name ='Steven Spielberg');
```

Question-13

```
ALTER TABLE Movie_cast
ADD Coloumn is_lead Boolean NOT NULL DEFAULT =0;

UPDATE Movie_Cast
SET is_lead =1 Where
cas_role IN('hero','heroine');
```

Question-14

```
SELECT * FROM Movie
Where mov_type ='Kids';
```

Question-15

```
SELECT * FROM Movie
Where mov_type ='Animated' AND mov_year >2022
AND mov_id IN(SELECT mov_id FROM Rating
WHERE
     rev_stars >=3.5);
```

Question-16

```
SELECT * FROM Movie
WHERE
    mov_type = 'Action';
```

Crete database

```
mysql> CREATE DATABASE IF NOT EXISTS Assignment_02;
Query OK, 1 row affected (0.08 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| Assignment_02      |
| Books_shop         |
| Shirts_db          |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| x                  |
+--------------------+
8 rows in set (0.02 sec)
```

```
mysql> USE Assignment_02;
Database changed
mysql> SELECT DATABASE();
+---------------+
| DATABASE()    |
+---------------+
| Assignment_02 |
+---------------+
1 row in set (0.00 sec)
```

Question - 01

Creating tables in database

```
mysql> CREATE TABLE Actor (
    ->     act_id INT NOT NULL AUTO_INCREMENT,
    ->     act_name VARCHAR(50) NOT NULL,
    ->     act_gender CHAR(1) NOT NULL CHECK (act_gender IN ('M' , 'F')),
    ->     PRIMARY KEY (act_id)
    -> );
Query OK, 0 rows affected (0.12 sec)

mysql> desc Actor;
+------------+-------------+------+-----+---------+----------------+
| Field      | Type        | Null | Key | Default | Extra          |
+------------+-------------+------+-----+---------+----------------+
| act_id     | int         | NO   | PRI | NULL    | auto_increment |
| act_name   | varchar(50) | NO   |     | NULL    |                |
| act_gender | char(1)     | NO   |     | NULL    |                |
+------------+-------------+------+-----+---------+----------------+
3 rows in set (0.02 sec)
```

```
mysql> CREATE TABLE Director (
    ->     dir_id INT NOT NULL AUTO_INCREMENT,
    ->     dir_name VARCHAR(50) NOT NULL,
    ->     dir_phone VARCHAR(15) UNIQUE,
    ->     PRIMARY KEY (dir_id)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> desc Director;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| dir_id    | int         | NO   | PRI | NULL    | auto_increment |
| dir_name  | varchar(50) | NO   |     | NULL    |                |
| dir_phone | varchar(15) | YES  | UNI | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
3 rows in set (0.01 sec)
```

```
mysql> CREATE TABLE Movie (
    ->     mov_id INT NOT NULL AUTO_INCREMENT,
    ->     mov_title VARCHAR(50) NOT NULL,
    ->     mov_year INT NOT NULL CHECK (mov_year >= 1900 AND mov_year <= 2030),
    ->     mov_lang VARCHAR(20) NOT NULL DEFAULT 'English',
    ->     mov_type VARCHAR(20),
    ->     dir_id INT NOT NULL,
    ->     PRIMARY KEY (mov_id),
    ->     CONSTRAINT fk_movie_director FOREIGN KEY (dir_id)
    ->         REFERENCES Director (dir_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> desc Movie;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| mov_id    | int         | NO   | PRI | NULL    | auto_increment |
| mov_title | varchar(50) | NO   |     | NULL    |                |
| mov_year  | int         | NO   |     | NULL    |                |
| mov_lang  | varchar(20) | NO   |     | English |                |
| mov_type  | varchar(20) | YES  |     | NULL    |                |
| dir_id    | int         | NO   | MUL | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
```

```
mysql> CREATE TABLE Movie_Cast (
    ->     cast_role VARCHAR(50) NOT NULL, act_id INT NOT NULL, mov_id INT NOT NULL,
    ->     PRIMARY KEY (act_id , mov_id , cast_role),
    ->     CONSTRAINT fk_cast_actor FOREIGN KEY (act_id) REFERENCES Actor (act_id) ON DELETE CASCADE O
N UPDATE CASCADE,
    ->     CONSTRAINT fk_cast_movie FOREIGN KEY (mov_id) REFERENCES Movie (mov_id) ON DELETE CASCADE O
N UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> desc Movie_Cast;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| cast_role | varchar(50) | NO   | PRI | NULL    |       |
| act_id    | int         | NO   | PRI | NULL    |       |
| mov_id    | int         | NO   | PRI | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
3 rows in set (0.01 sec)
```

```
mysql> CREATE TABLE Rating (
    ->     rating_id INT AUTO_INCREMENT, mov_id INT,
    ->     rev_stars DECIMAL(2 , 1 ) CHECK(rev_stars >= 0 AND rev_stars <= 5.0),
    ->     PRIMARY KEY (rating_id),
    ->     CONSTRAINT fk_rating_movie FOREIGN KEY (mov_id) REFERENCES Movie (mov_id) ON DELETE CASCADE
 ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> desc Rating;
+-----------+--------------+------+-----+---------+----------------+
| Field     | Type         | Null | Key | Default | Extra          |
+-----------+--------------+------+-----+---------+----------------+
| rating_id | int          | NO   | PRI | NULL    | auto_increment |
| mov_id    | int          | YES  | MUL | NULL    |                |
| rev_stars | decimal(2,1) | YES  |     | NULL    |                |
+-----------+--------------+------+-----+---------+----------------+
3 rows in set (0.01 sec)
```

```
mysql> show tables;
+-------------------------+
| Tables_in_Assignment_02 |
+-------------------------+
| Actor                   |
| Director                |
| Movie                   |
| Movie_Cast              |
| Rating                  |
+-------------------------+
5 rows in set (0.02 sec)
```

Question - 02

After creating tables inserting data

```
mysql> select * from Actor;
+--------+--------------------+------------+
| act_id | act_name           | act_gender |
+--------+--------------------+------------+
|      1 | Cary Grant         | M          |
|      2 | Grace Kelly        | F          |
|      3 | James Stewart      | M          |
|      4 | Kim Novak          | F          |
|      5 | Tippi Hedren       | F          |
|      6 | Tom Hanks          | M          |
|      7 | Meryl Streep       | F          |
|      8 | Harrison Ford      | M          |
|      9 | Meg Ryan           | F          |
|     10 | Richard Dreyfuss   | M          |
|     11 | Denzel Washington  | M          |
|     12 | Sandra Bullock     | F          |
|     13 | Brad Pitt          | M          |
|     14 | Angelina Jolie     | F          |
|     15 | Leonardo DiCaprio  | M          |
|     16 | Kate Winslet       | F          |
|     17 | Johnny Depp        | M          |
|     18 | Helena Carter      | F          |
|     19 | Will Smith         | M          |
|     20 | Cameron Diaz       | F          |
|     21 | Keanu Reeves       | M          |
|     22 | Jennifer Aniston   | F          |
|     23 | Robert Downey Jr.  | M          |
|     24 | Scarlett Johansson | F          |
|     25 | Chris Evans        | M          |
|     26 | Cate Blanchett     | F          |
|     27 | Sam Neill          | M          |
|     28 | Laura Dern         | F          |
|     29 | Joseph Gordon      | M          |
|     30 | Embeth Dave        | F          |
+--------+--------------------+------------+
30 rows in set (0.01 sec)
```

```
mysql> select * from Director;
+--------+-------------------+--------------+
| dir_id | dir_name          | dir_phone    |
+--------+-------------------+--------------+
|      1 | Alfred Hitchcock  | 0300-1000001 |
|      2 | Steven Spielberg  | 0300-1000002 |
|      3 | Martin Scorsese   | 0300-1000003 |
|      4 | Quentin Tarantino | 0300-1000004 |
|      5 | Christopher Nolan | 0300-1000005 |
|      6 | James Cameron     | 0300-1000006 |
|      7 | Ridley Scott      | 0300-1000007 |
|      8 | David Fincher     | 0300-1000008 |
|      9 | Denis Villeneuve  | 0300-1000009 |
|     10 | Wes Anderson      | 0300-1000010 |
|     11 | Joel Coen         | 0300-1000011 |
|     12 | Ethan Coen        | 0300-1000012 |
|     13 | Paul Thomas       | 0300-1000013 |
|     14 | Darren Aronofsky  | 0300-1000014 |
|     15 | Duncan Jones      | 0300-1000015 |
|     16 | Michel Gondry     | 0300-1000016 |
|     17 | Charlie Kaufman   | 0300-1000017 |
|     18 | Tony Scott        | 0300-1000018 |
|     19 | Michael Mann      | 0300-1000019 |
|     20 | Spike Lee         | 0300-1000020 |
|     21 | Kathryn Bigelow   | 0300-1000021 |
|     22 | Peter Jackson     | 0300-1000022 |
|     23 | Francis Coppola   | 0300-1000023 |
|     24 | Stanley Kubrick   | 0300-1000024 |
|     25 | George Lucas      | 0300-1000025 |
|     26 | Satyajit Ray      | 0300-1000026 |
|     27 | Akira Kurosawa    | 0300-1000027 |
|     28 | Roman Polanski    | 0300-1000028 |
|     29 | Elia Kazan        | 0300-1000029 |
|     30 | Billy Wilder      | 0300-1000030 |
+--------+-------------------+--------------+
30 rows in set (0.01 sec)
```

```
mysql> select * from Movie;
+--------+-----------------------------+----------+----------+-----------+--------+
| mov_id | mov_title                   | mov_year | mov_lang | mov_type  | dir_id |
+--------+-----------------------------+----------+----------+-----------+--------+
|      1 | Vertigo                     |     1958 | English  | Thriller  |      1 |
|      2 | Rear Window                 |     1954 | English  | Thriller  |      1 |
|      3 | Psycho                      |     1960 | English  | Horror    |      1 |
|      4 | The Birds                   |     1963 | English  | Horror    |      1 |
|      5 | Rope                        |     1948 | English  | Thriller  |      1 |
|      6 | Jaws                        |     1975 | English  | Thriller  |      2 |
|      7 | Jurassic Park               |     1993 | English  | Adventure |      2 |
|      8 | E.T. the Extra-Terrestrial  |     1982 | English  | Sci-Fi    |      2 |
|      9 | War Horse                   |     2011 | English  | Drama     |      2 |
|     10 | Saving Private Ryan         |     1998 | English  | War       |      2 |
|     11 | Taxi Driver                 |     1976 | English  | Drama     |      3 |
|     12 | Raging Bull                 |     1980 | English  | Drama     |      3 |
|     13 | The Irishman                |     2019 | English  | Crime     |      3 |
|     14 | Goodfellas                  |     1990 | English  | Crime     |      3 |
|     15 | The Wolf of Wall Street     |     2013 | English  | Drama     |      3 |
|     16 | Pulp Fiction                |     1994 | English  | Crime     |      4 |
|     17 | Kill Bill Vol. 1            |     2003 | English  | Action    |      4 |
|     18 | Inglourious Basterds        |     2009 | English  | War       |      4 |
|     19 | The Hateful Eight           |     2015 | English  | Thriller  |      4 |
|     20 | Django Unchained            |     2012 | English  | Western   |      4 |
|     21 | Inception                   |     2010 | English  | Sci-Fi    |      5 |
|     22 | Interstellar                |     2014 | English  | Sci-Fi    |      5 |
|     23 | The Dark Knight             |     2008 | English  | Action    |      5 |
|     24 | Titanic                     |     1997 | English  | Romance   |      6 |
|     25 | Avatar                      |     2009 | English  | Sci-Fi    |      6 |
|     26 | Blade Runner 2049           |     2017 | English  | Sci-Fi    |      7 |
|     27 | Gladiator                   |     2000 | English  | Action    |      7 |
|     28 | The Social Network          |     2010 | English  | Drama     |      8 |
|     29 | Gone Girl                   |     2014 | English  | Thriller  |      8 |
|     30 | Dune                        |     2021 | English  | Sci-Fi    |      9 |
+--------+-----------------------------+----------+----------+-----------+--------+
30 rows in set (0.00 sec)
```

```
mysql> select * from Movie_Cast;
+-----------+--------+--------+
| cast_role | act_id | mov_id |
+-----------+--------+--------+
| hero      |      1 |      1 |
| heroine   |      2 |      1 |
| brother   |      3 |      1 |
| friend    |      4 |      1 |
| brother   |      1 |      2 |
| heroine   |      2 |      2 |
| hero      |      3 |      2 |
| friend    |      5 |      2 |
| hero      |      1 |      3 |
| heroine   |      2 |      3 |
| brother   |      3 |      3 |
| friend    |      4 |      3 |
| brother   |      1 |      4 |
| heroine   |      2 |      4 |
| friend    |      3 |      4 |
| hero      |      5 |      4 |
| friend    |      1 |      5 |
| heroine   |      2 |      5 |
| hero      |      3 |      5 |
| brother   |      4 |      5 |
| brother   |      6 |      6 |
| friend    |      7 |      6 |
| heroine   |      9 |      6 |
| hero      |     10 |      6 |
| hero      |      8 |      7 |
| heroine   |     26 |      7 |
| brother   |     27 |      7 |
| friend    |     28 |      7 |
| hero      |      6 |      8 |
| brother   |     11 |      8 |
| heroine   |     12 |      8 |
| friend    |     13 |      8 |
+-----------+--------+--------+
32 rows in set (0.00 sec)
```

```
mysql> select * from Rating;
+-----------+--------+-----------+
| rating_id | mov_id | rev_stars |
+-----------+--------+-----------+
|         1 |      1 |       4.5 |
|         2 |      1 |       5.0 |
|         3 |      1 |       4.8 |
|         4 |      2 |       4.7 |
|         5 |      2 |       4.5 |
|         6 |      3 |       4.8 |
|         7 |      3 |       4.6 |
|         8 |      4 |       4.5 |
|         9 |      4 |       4.7 |
|        10 |      5 |       4.3 |
|        11 |      6 |       4.5 |
|        12 |      6 |       4.6 |
|        13 |      7 |       4.4 |
|        14 |      7 |       4.5 |
|        15 |      8 |       4.8 |
|        16 |      8 |       4.7 |
|        17 |      8 |       4.5 |
|        18 |      9 |       4.2 |
|        19 |     10 |       4.6 |
|        20 |     10 |       4.7 |
|        21 |     11 |       4.5 |
|        22 |     12 |       4.7 |
|        23 |     13 |       4.3 |
|        24 |     14 |       4.8 |
|        25 |     15 |       4.6 |
|        26 |     20 |       4.9 |
|        27 |     21 |       4.8 |
|        28 |     22 |       4.5 |
|        29 |     28 |       4.7 |
|        30 |     30 |       4.6 |
+-----------+--------+-----------+
30 rows in set (0.00 sec)
```

Question - 03

```
mysql> SELECT mov_title
    -> FROM Movie
    -> WHERE dir_id = (SELECT dir_id FROM Director WHERE dir_name = 'Alfred Hitchcock');
+-------------+
| mov_title   |
+-------------+
| Vertigo     |
| Rear Window |
| Psycho      |
| The Birds   |
| Rope        |
+-------------+
5 rows in set (0.01 sec)
```

Question - 04

```
mysql> SELECT mov_title, act_name FROM Movie, Actor
    -> WHERE act_id IN (SELECT act_id FROM Movie_Cast WHERE mov_id = Movie.mov_id);
+----------------------------+-------------------+
| mov_title                  | act_name          |
+----------------------------+-------------------+
| Vertigo                    | Cary Grant        |
| Vertigo                    | Grace Kelly       |
| Vertigo                    | James Stewart     |
| Vertigo                    | Kim Novak         |
| Rear Window                | Cary Grant        |
| Rear Window                | Grace Kelly       |
| Rear Window                | James Stewart     |
| Rear Window                | Tippi Hedren      |
| Psycho                     | Cary Grant        |
| Psycho                     | Grace Kelly       |
| Psycho                     | James Stewart     |
| Psycho                     | Kim Novak         |
| The Birds                  | Cary Grant        |
| The Birds                  | Grace Kelly       |
| The Birds                  | James Stewart     |
| The Birds                  | Tippi Hedren      |
| Rope                       | Cary Grant        |
| Rope                       | Grace Kelly       |
| Rope                       | James Stewart     |
| Rope                       | Kim Novak         |
| Jaws                       | Tom Hanks         |
| Jaws                       | Meryl Streep      |
| Jaws                       | Meg Ryan          |
| Jaws                       | Richard Dreyfuss  |
| Jurassic Park              | Harrison Ford     |
| Jurassic Park              | Cate Blanchett    |
| Jurassic Park              | Sam Neill         |
| Jurassic Park              | Laura Dern        |
| E.T. the Extra-Terrestrial | Tom Hanks         |
| E.T. the Extra-Terrestrial | Denzel Washington |
| E.T. the Extra-Terrestrial | Sandra Bullock    |
| E.T. the Extra-Terrestrial | Brad Pitt         |
+----------------------------+-------------------+
```

Question - 05

```
mysql> SELECT COUNT(*) AS brother_count FROM Movie_Cast
    -> WHERE cast_role = 'brother';
+---------------+
| brother_count |
+---------------+
|             8 |
+---------------+
1 row in set (0.00 sec)
```

Question - 06

```
mysql> SELECT mov_title FROM Movie
    -> WHERE mov_id IN (SELECT mov_id FROM Movie_Cast WHERE act_id IN (SELECT act_id FROM Movie_Cast GROUP BY act_id HAVING COUNT(mov_id) >= 2));
+----------------------------+
| mov_title                  |
+----------------------------+
| Vertigo                    |
| Rear Window                |
| Psycho                     |
| The Birds                  |
| Rope                       |
| Jaws                       |
| E.T. the Extra-Terrestrial |
+----------------------------+
7 rows in set (0.00 sec)
```

Question - 07

```
mysql> SELECT act_name
    -> FROM Actor
    -> WHERE
    -> act_id IN
    ->     (SELECT act_id FROM Movie_Cast WHERE mov_id IN
    ->         (SELECT mov_id FROM Movie WHERE mov_year < 2021))
    -> AND
    -> act_id IN
    ->     (SELECT act_id FROM Movie_Cast WHERE mov_id IN
    ->         (SELECT mov_id FROM Movie WHERE mov_year > 2023));
+------------+
| act_name   |
+------------+
| Tom Cruise |
+------------+
1 row in set (0.01 sec)
```

Question - 08

```
mysql> SELECT mov_title
    -> FROM Movie
    -> WHERE
    -> mov_id IN (SELECT mov_id FROM Movie_Cast WHERE act_id IN (SELECT act_id FROM Actor WHERE act_name IN (SELECT dir_name FROM Director)));
+-----------+
| mov_title |
+-----------+
| Hello     |
+-----------+
1 row in set (0.00 sec)
```

Question - 09

```
mysql> SELECT dir_name
    -> FROM Director
    -> WHERE
    ->      dir_id IN (SELECT dir_id FROM Movie WHERE mov_year >= 2020 AND mov_year <= 2024
    -> GROUP BY dir_id HAVING COUNT(mov_id) > 5);
Empty set (0.00 sec)
```

Question - 10

```
mysql> SELECT mov_title, (SELECT MAX(rev_stars) FROM Rating WHERE Rating.mov_id = Movie.mov_id) AS highest_stars FROM Movie
    -> WHERE mov_id IN (SELECT mov_id FROM Rating) ORDER BY mov_title;
+----------------------------+---------------+
| mov_title                  | highest_stars |
+----------------------------+---------------+
| Django Unchained           |           4.9 |
| Dune                       |           4.6 |
| E.T. the Extra-Terrestrial |           4.8 |
| Goodfellas                 |           4.8 |
| Inception                  |           4.8 |
| Interstellar               |           4.5 |
| Jaws                       |           4.6 |
| Jurassic Park              |           4.5 |
| New Movie                  |           4.8 |
| Old Movie                  |           4.6 |
| Psycho                     |           4.8 |
| Raging Bull                |           4.7 |
| Rear Window                |           4.7 |
| Rope                       |           4.3 |
| Saving Private Ryan        |           4.7 |
| Taxi Driver                |           4.5 |
| The Birds                  |           4.7 |
| The Irishman               |           4.3 |
| The Social Network         |           4.7 |
| The Wolf of Wall Street    |           4.6 |
| Vertigo                    |           5.0 |
| War Horse                  |           4.2 |
+----------------------------+---------------+
22 rows in set (0.01 sec)
```

Question - 11

```
mysql> SELECT (SELECT mov_title FROM Movie WHERE Movie.mov_id = Rating.mov_id) AS mov_title, AVG(rev_stars) AS avg_rating
    -> FROM Rating GROUP BY mov_id;
+----------------------------+------------+
| mov_title                  | avg_rating |
+----------------------------+------------+
| Vertigo                    |    4.76667 |
| Rear Window                |    4.60000 |
| Psycho                     |    4.70000 |
| The Birds                  |    4.60000 |
| Rope                       |    4.30000 |
| Jaws                       |    4.55000 |
| Jurassic Park              |    4.45000 |
| E.T. the Extra-Terrestrial |    4.66667 |
| War Horse                  |    4.20000 |
| Saving Private Ryan        |    4.65000 |
| Taxi Driver                |    4.50000 |
| Raging Bull                |    4.70000 |
| The Irishman               |    4.30000 |
| Goodfellas                 |    4.80000 |
| The Wolf of Wall Street    |    4.60000 |
| Django Unchained           |    4.90000 |
| Inception                  |    4.80000 |
| Interstellar               |    4.50000 |
| The Social Network         |    4.70000 |
| Dune                       |    4.60000 |
| Old Movie                  |    4.60000 |
| New Movie                  |    4.80000 |
+----------------------------+------------+
22 rows in set (0.00 sec)
```

Question - 12

```
mysql> SELECT mov_title
    -> FROM Movie
    -> WHERE dir_id = (SELECT dir_id FROM Director WHERE dir_name = 'Steven Spielberg');
+----------------------------+
| mov_title                  |
+----------------------------+
| Jaws                       |
| Jurassic Park              |
| E.T. the Extra-Terrestrial |
| War Horse                  |
| Saving Private Ryan        |
| Top Gun                    |
+----------------------------+
6 rows in set (0.00 sec)
```

Question - 13

```
mysql> ALTER TABLE Movie_Cast
    -> ADD COLUMN is_lead BOOLEAN NOT NULL DEFAULT 0;
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Movie_Cast
    -> SET is_lead = 1 WHERE cast_role IN ('hero', 'heroine');
Query OK, 22 rows affected (0.02 sec)
Rows matched: 22  Changed: 22  Warnings: 0

mysql>
```

Question - 14

```
mysql> SELECT * FROM Movie WHERE mov_type = 'Kids';
+--------+-----------+----------+----------+----------+--------+
| mov_id | mov_title | mov_year | mov_lang | mov_type | dir_id |
+--------+-----------+----------+----------+----------+--------+
|     36 | Toy Story |     1995 | English  | Kids     |      2 |
+--------+-----------+----------+----------+----------+--------+
1 row in set (0.01 sec)
```

Question - 15

```
mysql> SELECT * FROM Movie
    -> WHERE mov_type = 'Animated' AND mov_year > 2022 AND mov_id IN (SELECT mov_id FROM Rating WHERE rev_stars >= 3.5);
+--------+------------------+----------+----------+----------+--------+
| mov_id | mov_title        | mov_year | mov_lang | mov_type | dir_id |
+--------+------------------+----------+----------+----------+--------+
|     37 | Super Kids       |     2023 | English  | Animated |      2 |
|     38 | Magic Forest     |     2024 | English  | Animated |      3 |
|     39 | Robot Adventures |     2025 | English  | Animated |      4 |
+--------+------------------+----------+----------+----------+--------+
3 rows in set (0.00 sec)
```

Question - 16

```
mysql> SELECT * FROM Movie
    -> WHERE mov_type = 'Action';
+--------+-----------------+----------+----------+----------+--------+
| mov_id | mov_title       | mov_year | mov_lang | mov_type | dir_id |
+--------+-----------------+----------+----------+----------+--------+
|     17 | Kill Bill Vol. 1 |     2003 | English  | Action   |      4 |
|     23 | The Dark Knight |     2008 | English  | Action   |      5 |
|     27 | Gladiator       |     2000 | English  | Action   |      7 |
|     32 | New Movie       |     2024 | English  | Action   |     24 |
|     33 | Top Gun         |     1986 | English  | Action   |      2 |
|     35 | Hello           |     1986 | English  | Action   |      1 |
+--------+-----------------+----------+----------+----------+--------+
6 rows in set (0.00 sec)
```